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Does Tax-Benefit Linkage Matter for the Incidence of Payroll Taxes?

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1 Big picture

- **Question.** Do payroll taxes get shifted to workers in the form of lower earnings, and does the answer depend on whether contributions are transparently linked to future social-insurance benefits?
- **Main idea.** Payroll taxes are not all the same. A contribution that buys an actuarially meaningful pension benefit is economically closer to deferred compensation; a contribution used for general spending is closer to a pure tax on labour. The paper asks whether this distinction matters empirically.
- **Setting.** France is useful because it has a long administrative earnings panel, a rich payroll-tax schedule with thresholds, and six large reforms over several decades. Some reforms changed payroll taxes tied to pension benefits, while others changed payroll or income taxes with no meaningful tax-benefit link.
- **Reforms studied.** The paper studies two payroll tax reforms with strong tax-benefit linkage, two employer payroll tax reforms without linkage, and two income-tax reforms that resemble payroll taxes but have no linkage.
- **Empirical design.** For threshold-based reforms, the authors compare workers just above and just below the relevant earnings threshold before and after the reform. For the profit-sharing income-tax reform, they compare workers in firms with versus without profit-sharing exposure.
- **Main result with linkage.** When payroll tax increases are clearly tied to expected benefits, the tax is fully passed through to workers in the medium run. This is true whether the tax is statutorily called an employer contribution or an employee contribution.
- **Main result without linkage.** When payroll taxes are not tied to future benefits, employer-side payroll tax changes are not substantially shifted to the directly affected workers. The individual-level burden lines up with statutory incidence.
- **Income-tax comparison.** A worker-side income-tax increase, which lacks tax-benefit linkage but is statutorily borne by employees, is shifted to workers through lower net earnings. This supports the idea that, without linkage, statutory incidence matters at the individual level.
- **Meta-analysis.** The paper adds a literature meta-analysis. Employer payroll tax reforms with strong linkage show average pass-through near 100 percent, while employer payroll tax reforms without linkage show low average pass-through.
- **Policy message.** Tax-benefit linkage can make payroll taxes less distortionary and more likely to be accepted as deferred compensation. Ignoring linkage can lead to wrong conclusions about who bears payroll taxes and how payroll-tax reforms redistribute resources.

2 Section-by-section detailed notes

2.1 Section 1: Introduction

2.1.1 Motivation

Payroll taxes are a major revenue source in OECD countries and especially in countries with large social insurance systems. They are often split into employer and employee components, creating a classic statutory-incidence problem: the law says who remits or is charged the tax, but economic theory asks who ultimately bears the burden.

The standard textbook model emphasizes supply and demand elasticities and predicts that statutory incidence should not matter. If labour supply is inelastic relative to labour demand, payroll taxes should largely be borne by workers. Yet the empirical literature is mixed. Some studies find near-complete wage pass-through, while others find little pass-through.

The paper’s proposed reconciliation is **tax-benefit linkage**: the degree to which paying the tax increases a worker’s future benefits. This design feature differs across payroll taxes and across countries. A payroll contribution to an actuarially fair pension scheme is not perceived like a tax; it resembles forced saving or deferred compensation. By contrast, a payroll tax financing unrelated programmes may be perceived like a pure tax.

2.1.2 Research contribution

The paper contributes by combining six French reforms in one empirical framework. This creates variation along three dimensions:

- whether the tax is linked to benefits;
- whether the statutory incidence is on employers or employees;
- whether the reform is a payroll tax or a labour-income tax resembling a payroll tax.

The main empirical outcome is the effect of tax changes on earnings. The core object is pass-through to workers: if a tax increase raises labour cost but posted earnings fall by the same amount, workers bear the tax. If posted earnings do not adjust, employers bear the tax at the individual level.

2.1.3 Headline results

The introduction reports three central findings:

- **Linked payroll taxes:** increases in employer and employee payroll taxes tied to future benefits are fully passed through to affected workers within roughly five to six years.
- **Unlinked employer taxes:** employer payroll tax increases without linkage and a payroll-based corporate income tax credit have little effect on directly affected workers’ earnings.
- **Unlinked employee-side tax:** an income tax on workers is borne by workers through lower net earnings.

Interpretation: The paper is not merely estimating one incidence parameter. It is showing that incidence depends on the institutional meaning of the tax. The same broad category—“payroll tax”—can have different incidence depending on whether workers see the payment as buying a benefit.

2.2 Section 2: Conceptual framework

2.2.1 General setup and earnings concepts

The benchmark model is a competitive labour-market incidence model with two labour inputs: treated workers and control workers. Treated workers are affected by a reform; control workers are not. Let

$$L_T, L_C$$

be the two labour inputs and let the production function be

$$F(L_T, L_C),$$

homogeneous of degree one with positive and diminishing marginal products. The elasticity of substitution between treated and control workers is denoted by σ .

Let w_k be posted earnings for worker type $k \in \{T, C\}$. If payroll taxes nominally paid by firms apply at rate τ_k , the labour cost is

$$z_k \equiv w_k(1 + \tau_k).$$

Interpretation: The distinction between posted wage and labour cost is essential. The empirical question is whether a change in τ_k changes w_k , z_k , or both.

2.2.2 Tax-benefit linkage

Workers may value the future benefits financed by payroll taxes. The perceived wage is

$$\tilde{w}_k \equiv w_k(1 + q\tau_k),$$

where $q \in [0, 1]$ captures perceived tax-benefit linkage.

- $q = 0$: workers treat the payroll tax as a pure tax and do not value the associated benefit.
- $q = 1$: workers fully value the benefit; the tax is perceived as actuarially fair deferred compensation.
- Intermediate q : the tax has partial perceived linkage.

Labour supply is written as

$$L_k^S = L_k^S(\tilde{w}_k, R_k) = L_k^S(w_k(1 + q\tau_k), R_k),$$

where R_k is non-labour income. The labour-supply elasticity with respect to posted wage is denoted by η_k^S .

Interpretation: This reduced-form modelling of linkage is the bridge between public-finance theory and the French institutional reforms. The paper does not need to model every pension rule; it needs to distinguish reforms in which workers can plausibly see the tax as tied to future benefits.

2.2.3 Pass-through formula

The pass-through rate to workers is

$$\rho = -\frac{d \ln(w_T/w_C)}{d \ln(1 + \tau_T)}.$$

Assuming similar labour supply elasticities for treated and control workers, the paper's benchmark model gives

$$\rho \approx \frac{\sigma + \eta^S q}{\sigma + \eta^S}.$$

Read:

- The numerator includes σ , the ability of firms to substitute between treated and control workers.
- The numerator also includes $\eta^S q$, the part of labour supply that is neutralized because workers value the benefit.
- The denominator includes total labour-market adjustment forces, $\sigma + \eta^S$.

Economic message: If $q = 1$, pass-through is approximately one regardless of elasticities. If $q = 0$, pass-through depends on the relative sizes of substitution and labour-supply elasticities. This is why tax-benefit linkage can dominate the standard incidence logic.

2.2.4 Three polar cases

The paper highlights three cases:

1. **Full linkage:** if $q = 1$, payroll taxes are fully shifted to workers and quantities need not change. Workers are effectively receiving future benefits in exchange for lower current wages.
2. **No linkage and high substitution:** if $q = 0$ and $\sigma \gg \eta^S$, taxes are shifted to workers. This is the familiar case in which treated workers cannot maintain wages because firms can substitute toward control workers.
3. **No linkage and low substitution:** if $q = 0$ and $\eta^S \gg \sigma$, there is little pass-through to wages, and labour costs absorb the tax increase.

Interpretation: The empirical results strongly support the first case for linked payroll taxes. For unlinked taxes, the results look more statutory than the competitive model would usually predict, motivating the discussion of wage bargaining, fairness norms, and perception effects.

2.3 Section 3: Tax reforms

2.3.1 French earnings taxation

French payroll taxes finance health care, pensions, unemployment insurance, child benefits, and other social programmes. Although many are earmarked, not all are equally linked to individual benefits. The paper distinguishes:

- taxes with no individual benefit linkage, such as health or family contributions;
- taxes with indirect linkage, such as some earnings-related social insurance contributions;
- taxes with strong linkage, especially complementary pension contributions where benefits depend directly on contributions.

The French payroll tax schedule is bracketed around thresholds, especially the Social Security threshold (SST). This creates useful quasi-experimental variation: workers above and below a threshold face different changes in marginal and average tax rates after reforms.

2.3.2 Linked payroll tax reforms: L1 and L2

L1 is an increase in complementary pension payroll taxes for non-executive workers affiliated with the ARRCO scheme. It affected taxes around the SST in the early 2000s and increased both employer and employee contributions. The pension system is point-based, so contributions are transparently tied to future benefit entitlements.

L2 is an increase in complementary pension payroll taxes for executive workers affiliated with the AGIRC scheme. It occurs around a higher threshold, roughly four times the SST, and also has strong tax-benefit linkage through pension points.

Interpretation: In these reforms, workers above the relevant threshold pay more, but they also receive higher expected pension benefits. The model predicts full pass-through to workers and limited quantity responses.

2.3.3 Unlinked employer payroll tax reforms: NL1 and NL2

NL1 removes or reduces an employer payroll tax related to family benefits around the SST. The contribution finances benefits that are not tied to individual earnings or contributions.

NL2 modifies employer health-care payroll taxes around the SST. Health benefits are not tied to a worker's individual payroll-tax contribution. Thus, the reform changes the firm's labour cost without changing the treated worker's expected individual benefits.

Interpretation: These reforms are crucial because they are employer-side and lack linkage. Standard models might still predict substantial pass-through to workers if substitution between treated and control workers is high. The paper finds little individual-level pass-through.

2.3.4 Income-tax reforms without linkage: NL3 and NL4

NL3 is a corporate income tax credit based on payroll expenses for workers below 2.5 times the minimum wage. Economically, it resembles a cut in employer payroll taxes for eligible workers, but it is not linked to worker benefits.

NL4 is an increase in a flat-rate income tax on earnings from profit-sharing bonuses. It resembles an employee-side payroll tax increase because it taxes labour earnings at the individual level, but it lacks any benefit linkage.

Interpretation: These two reforms help separate statutory incidence from tax-benefit linkage. If no-linkage taxes are borne according to statutory incidence, employer-side NL3 should not raise worker earnings much, while employee-side NL4 should lower net earnings.

2.4 Section 4: Data and tax measurement

2.4.1 Administrative earnings data

The paper uses French administrative matched employer–employee data. These data report annual earnings for wage earners over a long time span and allow the authors to follow workers around policy reforms. The analysis focuses on private-sector full-time wage earners with stable annual employment, which helps interpret earnings changes as wage incidence rather than shifts in hours or employment spells.

2.4.2 Tax calculation

The authors compute payroll taxes using a microsimulation tool based on detailed statutory payroll-tax schedules. This is necessary because French taxes depend on earnings brackets, worker type, social security thresholds, minimum wage multiples, and the exact reform year.

For each worker-year, the paper constructs:

- posted earnings w_{it} ;

- labour cost z_{it} ;
- average employer payroll tax rate $\tau_{it} = z_{it}/w_{it} - 1$;
- net earnings when employee-side taxes are included;
- tax changes induced by each reform.

Interpretation: The tax calculation is a first stage. The paper does not simply assign treatment status; it measures how much average tax rates actually changed for each treated group.

2.4.3 Treatment and control groups

For threshold reforms, treatment and control groups are based on pre-reform earnings relative to the relevant threshold.

- For reforms around one SST, treated workers are above the SST but not too far above it; control workers are slightly below the SST.
- For the reform around four SST, treated and control groups are high earners around the four-SST threshold.
- For the payroll-tax credit around 2.5 times the minimum wage, the treated group is below the eligibility cutoff and the control group is above it.
- For the profit-sharing income-tax reform, treatment is based on whether a worker's firm distributed profit-sharing before the reform.

Interpretation: The design is threshold-based but not a narrow regression discontinuity. It is a panel difference-in-differences design using workers near policy cutoffs and following them over time.

2.5 Section 5: Empirical approach

2.5.1 Baseline event-study first stage and reduced form

For employer payroll tax reforms, the first-stage equation is

$$\log(1 + \tau_{it}) = \alpha + \theta_i + \eta_t + \sum_{k=-m}^r \beta_k (T_i \times \mathbf{1}\{t = t_0 + k\}) + \varepsilon_{it}.$$

The reduced-form equation is

$$\log w_{it} = \mu + \nu_i + \pi_t + \sum_{k=-m}^r \gamma_k (T_i \times \mathbf{1}\{t = t_0 + k\}) + v_{it}.$$

- T_i is the treatment-group indicator.
- t_0 is the reference pre-reform year.
- Worker fixed effects absorb permanent earnings differences.
- Year fixed effects absorb aggregate wage trends.
- β_k measures the differential tax change.
- γ_k measures the differential earnings response.

Interpretation: The event-study form lets the reader check parallel pre-trends and observe the medium-run adjustment path.

2.5.2 Pass-through as a Wald ratio / 2SLS coefficient

The pass-through estimate in post-reform year k is

$$\hat{\rho}_k = -\frac{\hat{\gamma}_k}{\hat{\beta}_k}.$$

Equivalently, the paper estimates a 2SLS equation separately for each post-reform year:

$$\log w_{it} = \lambda + \phi_i + \omega_t - \rho_k \log(1 + \tau_{it}) + \sum_{\ell=-m, \ell \neq k}^r \delta_\ell (T_i \times \mathbf{1}\{t = t_0 + \ell\}) + \zeta_{it},$$

using the treatment-by-post-year interaction as the instrument for $\log(1 + \tau_{it})$.

Interpretation: A pass-through estimate of one means the tax is fully borne by workers. A pass-through estimate of zero means no individual-level wage adjustment relative to the control group.

2.5.3 Net earnings and employee-side taxes

For reforms affecting employee payroll taxes or income taxes, the authors modify the first stage to use a tax-inclusive measure that incorporates both employer and employee-side taxes, and they replace posted earnings with net earnings when appropriate.

Interpretation: This is important for reforms L1 and L2, where both employer and employee pension contributions rose, and for NL4, where the relevant tax is charged directly to employees through profit-sharing income.

2.5.4 Earnings versus hourly wages

Hours are not observed throughout the full period, so the paper uses total earnings. The authors restrict attention to full-time wage earners employed throughout the year. If hours responses existed, they would tend to bias some estimates toward finding more pass-through. The near-zero pass-through for unlinked employer taxes is therefore not easily explained by hours responses.

2.5.5 Identification assumptions

The core identifying assumption is that treated and control workers would have followed parallel earnings trends absent the reform. The paper supports this with:

- graphical pre-trend checks;
- worker fixed effects and year fixed effects;
- robustness to alternative treatment/control windows;
- worker-specific trends;
- doughnut-hole specifications removing workers near thresholds;
- nonparametric graphical evidence;
- placebo reforms in periods without tax changes.

2.6 Section 6: Results

2.6.1 Payroll tax reforms with linkage

For L1, average real posted earnings and labour costs move together after the pension payroll tax increase. The pass-through estimates are close to one. Table 3 reports last-year estimates around 1.01 and 1.00 for employer-side specifications, and around 0.91 and 1.20 when both employer and employee payroll taxes are included.

For L2, estimates are noisier because the treated group is high earners around four times the SST, but the pattern is again consistent with full pass-through. Table 3 reports last-year estimates above one, with large standard errors.

Interpretation: Linked payroll tax increases are borne by affected workers because workers receive corresponding pension benefit entitlements. The result is not sensitive to whether contributions are nominally employer or employee contributions.

2.6.2 Payroll tax reforms without linkage

For NL1, the pass-through estimates are close to zero. Table 4 reports a last-year estimate of 0.25 and a final-years estimate of -0.02.

For NL2, the last-year estimate is 0.44 and the final-years estimate is 0.41, but the authors emphasize that the evidence is affected by pre-trends and becomes smaller with trend controls. Overall, the result is not consistent with full shifting to directly affected workers.

Interpretation: Employer payroll tax increases without benefit linkage are not fully passed to the treated workers. At the individual level, the burden aligns more with statutory incidence: employer-side taxes are absorbed by employers rather than by directly affected workers' posted earnings.

2.6.3 Income tax reforms without linkage

For NL3, the corporate income tax credit based on payroll expenses resembles an employer payroll tax cut. Table 4 reports estimates near zero: last-year pass-through is -0.06 and final-years pass-through is -0.05. This means the employer-side tax credit did not substantially raise affected workers' earnings.

For NL4, the employee-side income tax increase lowers net earnings. Table 4 reports a last-year pass-through estimate of 0.89 and a final-years estimate of 0.75. This is consistent with the statutory burden falling on employees.

Interpretation: These income-tax reforms strengthen the statutory-incidence interpretation. Without tax-benefit linkage, employer-side taxes stay with employers at the individual level, and employee-side taxes are borne by workers.

2.6.4 Formal tests

Table 5 provides formal tests of the main hypotheses:

- For linked reforms, the hypothesis that incidence is on employers is rejected, while the hypothesis of full pass-through to workers is not rejected.
- For unlinked reforms, the hypothesis of statutory incidence is not rejected, while the opposite-incidence hypothesis is rejected.

- For employer-side reforms, pass-through differs significantly between linked and unlinked taxes.
- For employee-side reforms, pass-through does not differ significantly between linked and unlinked taxes because both are borne by workers.

2.7 Section 7: Interpretation and discussion

2.7.1 Meta-analysis

The paper compares its estimates with the broader literature. Figure 8 shows that employer payroll tax studies with strong linkage tend to find full pass-through to workers, while employer payroll tax studies without linkage tend to find low pass-through. The paper reports average pass-through of about 103 percent for employer payroll taxes with strong linkage and about 16 percent for those without linkage.

For employee-side payroll or income taxes, the average pass-through to workers is close to full and does not differ strongly by linkage.

Interpretation: The meta-analysis supports the paper’s argument that tax-benefit linkage helps reconcile previously conflicting results in the payroll-tax incidence literature.

2.7.2 Efficiency with linkage

When workers perceive payroll taxes as buying future benefits, payroll-tax increases should not reduce labour supply in the same way as pure taxes. The paper reports no evidence of differential exit rates from employment for linked reforms, consistent with the view that linked payroll taxes have lower deadweight costs.

2.7.3 Statutory incidence without linkage

The no-linkage results are harder to reconcile with a competitive model in which statutory incidence is irrelevant. The paper discusses several possible explanations:

- workers may perceive employer payroll taxes as firm taxes rather than their own taxes;
- wage bargaining and internal fairness norms may prevent firms from lowering wages only for treated workers;
- employer-side taxes may still be shifted at the firm or market level, which the worker-level design cannot fully observe.

Interpretation: The paper’s zero individual-level pass-through for unlinked employer taxes should not be read as a definitive statement that employers bear the entire aggregate burden. It says the directly affected workers do not bear the tax through lower relative earnings.

2.8 Section 8: Conclusion

The conclusion emphasizes that payroll tax incidence depends on institutional design. Linked payroll taxes are fully borne by workers because workers internalize the associated future benefits. Unlinked employer payroll taxes show little individual-level pass-through to workers. Employee-side taxes, whether payroll or income taxes, are borne by workers.

Takeaway: The paper’s policy message is that clear tax-benefit linkage can make payroll taxation more efficient and less distortionary. But for non-contributory payroll taxes, assuming automatic wage shifting can be misleading, especially when evaluating distributional consequences at the worker level.

3 Figures and tables

3.1 Figure 1 and Table 1: reform design and theoretical predictions

Read: Figure 1:

- Panels A and B show reforms around the Social Security threshold for non-executive workers: employer payroll rates and employee payroll rates differ above versus below the SST.
- Panels C and D show the higher-threshold reform for executives around four times the SST.
- Panel E shows the payroll-based corporate income tax credit below 2.5 times the minimum wage.
- Panel F shows the flat-rate employee tax reform on profit-sharing and base earnings.
- The visual structure makes clear that each reform creates differential tax variation around a threshold or exposure group.

Read: Table 1:

- The table classifies each reform by tax-benefit linkage, statutory incidence, and tax type.
- Linked reforms L1 and L2 are predicted to be fully passed to workers regardless of statutory side.
- Unlinked employer payroll taxes NL1 and NL2 have model-dependent predictions.
- NL3 resembles an employer payroll tax cut, while NL4 resembles an employee-side payroll tax increase.

Economic message: The paper’s empirical design is built around a clean conceptual taxonomy: linked versus unlinked, employer-side versus employee-side, payroll tax versus payroll-like income tax.

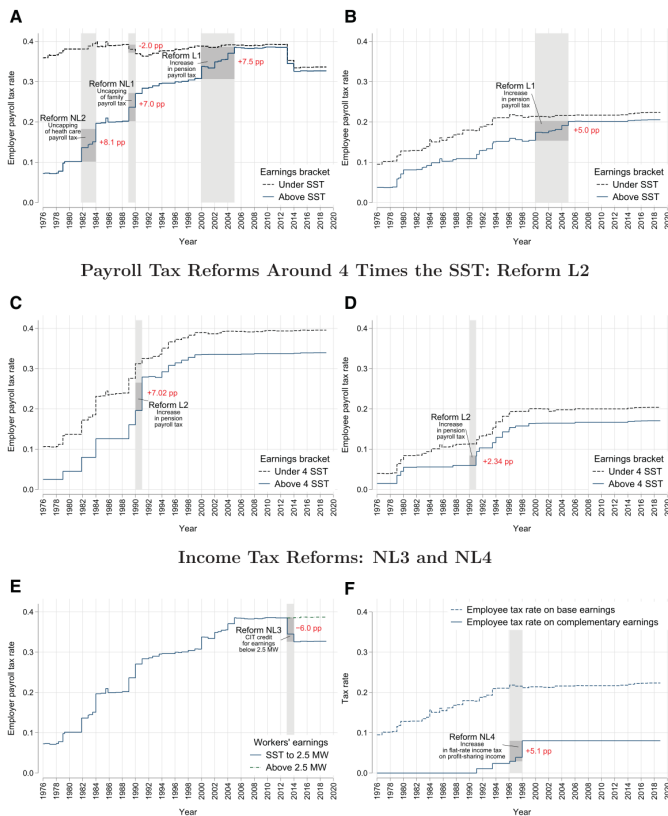


FIGURE 1

Payroll tax and income tax reforms: marginal tax rates above and below earnings thresholds, 1976–2019 (A) Employer payroll tax rates. (B) Employee payroll tax rates. (C) Employer payroll tax rates. (D) Employee payroll tax rates. (E) Employer tax credit below 2.5 MW (NL3) and (F) Employee tax rates by type of earnings (NL4)

Notes: Marginal payroll tax rates are expressed as a fraction of posted earnings. These rates are applied to different earnings brackets, defined with respect to the Social Security threshold (SST) or the Minimum Wage (MW). In (A) and (B) (Reforms L1, L2, and L3), the rates apply to non-executive workers, affiliated with the ARRCO complementary pension scheme, while in (C) and (D) (Reform L2), they apply to executive workers, affiliated with the AGIRC complementary pension scheme. In (E), Reform NL3 induced a cut in average tax rate, rather than a change in the marginal tax rate. In (F) (Reform NL4), profit-sharing income refers to earnings received by employees under mandatory profit-sharing schemes (*intéressement et participation*). Reform NL4 led to an increase in the CSG rate on earnings (inclusive of profit-sharing income), which was offset by a reduction in employee payroll taxes, which previously did not apply to profit-sharing income. Sources: *Institut des Politiques Publiques* (2023); TAXIPP 0.4.

(a) Figure 1: six reform tax schedules.

Tax-benefit linkage	Statutory incidence	Type of tax reform	Reform(s)	Predicted incidence	
				Wages	Quantities
Yes	Employers	Payroll tax	L1, L2	A broad class of models predict full pass-through to workers at the individual level	No effects on quantities if there is full pass-through to workers
	Employees	Payroll tax	L1, L2		
No	Employers	Payroll tax	NL1, NL2	Predicted wages and quantity responses depend on model and elasticities	
	Employees	Payroll tax (payroll-based corp. tax credit)	NL3		
	Employers	Payroll tax	[None]		
	Income tax	Income tax	NL4		

Notes: This table displays all possible combinations of tax-benefit linkage (2) × statutory incidence (2) × type of tax reform (2), alongside the reforms analysed in this article and the corresponding theoretical predictions about their incidence and employment effects (see Appendix A). By definition, income tax with linkage does not occur.

(b) Table 1: reform taxonomy and predictions.

3.2 Table 2: treatment and control groups

Read:

- Table 2 reports the baseline samples for each reform.
- For the threshold reforms, treatment and control groups are defined by earnings intervals in the reference year.
- For L1, NL1, and NL2, treatment is above one SST and control is slightly below one SST.
- For L2, treatment is above four SST and control is below four SST.
- For NL3, treatment is below 2.5 times the minimum wage because that is where the tax credit applies.
- For NL4, treatment is based on profit-sharing exposure rather than an earnings threshold.

Economic message: Table 2 is central for assessing credibility. It shows how close treated and control workers are in the earnings distribution and how the design differs across the six reforms.

TABLE 2
Summary statistics for the treatment and control groups

Sample:	Control group		Treatment group	
	Mean	S.D.	Mean	S.D.
Panel A. Reform L1: Increase in pension payroll tax for non-executives (2000–5)				
Reference year (t_0): 1999				
Posted earnings	0.9 to 1 SST		1 to 1.4 SST	
Percentile rank in the earnings distribution	[P64–P72]		[P72–P88]	
Age	41.4	(8.9)	43.1	(8.4)
Male	0.72	(0.45)	0.75	(0.43)
Annual posted earnings	30,078	(911)	36,457	(3,415)
Annual labour cost	43,366	(1,673)	52,276	(4,838)
Number of individuals	19,626		33,605	
Number of observations	183,022		314,511	
Panel B. Reform L2: Increase in pension payroll tax for executives (1991)				
Reference year (t_0): 1989				
Posted earnings	2.5 to 4 SST		4 to 5.5 SST	
Percentile rank in the earnings distribution	[P97–P100]		[P100]	
Age	44.6	(6.1)	46.9	(6.4)
Male	0.94	(0.23)	0.94	(0.24)
Annual posted earnings	84,092	(11,012)	125,608	(13,750)
Annual labour cost	114,270	(14,941)	167,338	(18,155)
Number of individuals	1,321		334	
Number of observations	10,568		2,672	
Panel C. Reform NL1: Uncapping of family payroll tax (1989 and 1990)				
Reference year (t_0): 1988				
Posted earnings	0.9 to 1 SST		1 to 1.4 SST	
Percentile rank in the earnings distribution	[P61–P69]		[P69–P86]	
Age	39.0	(9.3)	40.6	(8.8)
Male	0.74	(0.44)	0.77	(0.42)
Annual posted earnings	26,074	(789)	31,805	(3,025)
Annual labour cost	37,498	(1,363)	45,091	(3,953)
Number of individuals	24,390		46,579	
Number of observations	203,900		393,571	
Panel D. Reform NL2: Uncapping of health care payroll tax (1981 and 1983)				
Reference year (t_0): 1980				
Posted earnings	0.9 to 1 SST		1 to 1.4 SST	
Percentile rank in the earnings distribution	[P58–P67]		[P67–P86]	
Age	38.0	(10.7)	39.4	(10.1)
Male	0.76	(0.43)	0.79	(0.41)
Annual posted earnings	22,409	(683)	27,432	(2,627)
Annual labour cost	31,850	(1,041)	38,065	(3,080)
Number of individuals	30,988		63,134	
Number of observations	209,487		438,438	
Panel E. Reform NL3: Employer tax credit on payroll taxes (2013)				
Reference year (t_0): 2012				
Posted earnings	2.5 to 2.7 MW		2.3 to 2.5 MW	
Percentile rank in the earnings distribution	[P80–P84]		[P76–P80]	
Age	45.4	(9.1)	44.9	(9.3)
Male	0.75	(0.43)	0.74	(0.44)
Annual posted earnings	42,185	(939)	38,922	(934)
Annual labour cost	61,486	(2,061)	56,765	(1,992)
Number of individuals	13,374		20,448	
Number of observations	108,679		166,955	

(a) Table 2, main panels.

Sample:	Control group		Treatment group	
	Mean	S.D.	Mean	S.D.
Panel F. Reform NL4: Employee income tax hike on profit-sharing (1996 and 1997)				
Reference year (t_0): 1995				
Group	No profit-sharing [P1–P100]		Has profit-sharing [P1–P100]	
Percentile rank in the earnings distribution				
Age	38.9	(10)	39.2	(9.6)
Male	0.69	(0.46)	0.67	(0.47)
Annual posted earnings	20,469	(23,986)	30,246	(22,918)
Annual labour cost	28,942	(33,650)	42,418	(31,947)
Number of individuals	43,808		77,627	
Number of observations	153,975		410,145	

Notes: Each panel provides summary statistics for the sample used in the analysis of each reform. The statistics are presented for the reference year and for the baseline sample of full-time and full-year workers, separately for the treatment and control groups. Gross earnings and labour costs are expressed in 2010 euros. Standard deviations are shown in parentheses. Mean differences between the treatment and control groups are significant at the 5% level for all variables. Sources: DADS Panel; TAXIPP 0.4.

(b) Table 2, continued and notes.

3.3 Figures 2–3 and Table 3: linked payroll tax reforms

Read: Figures 2 and 3:

- Figure 2 shows earnings and labour costs for linked reforms L1 and L2.
- Before reform, treated and control trends are broadly aligned.
- After reform, labour costs rise for treated workers, and posted earnings adjust downward relative to the control group.
- Figure 3 converts these patterns into pass-through event studies.
- The pass-through estimates cluster around full shifting to workers, especially for L1.

Read: Table 3:

- L1 has last-year employer-side pass-through near 1.01 and controlled pass-through near 1.00.

- L1 also shows near-full pass-through when the full employer-plus-employee payroll tax change is considered.
- L2 estimates are noisier but consistent with full pass-through.
- The formal estimates are consistent with the conceptual prediction that linked contributions are borne by workers.

Economic message: Linked pension contributions behave like deferred compensation. Workers accept lower current earnings because the contributions increase expected pension benefits.

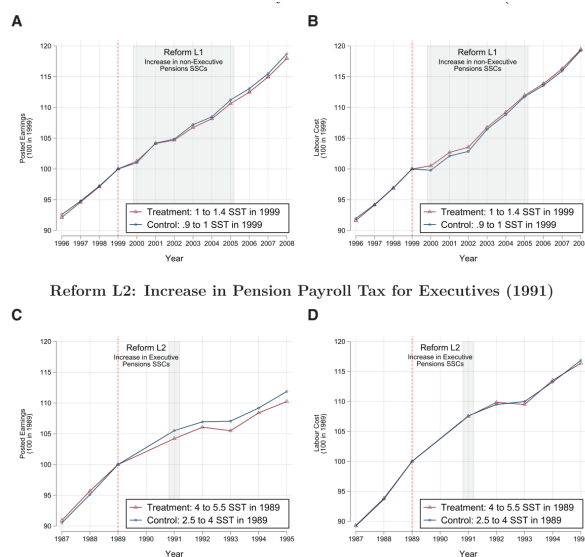


FIGURE 2

Earnings responses to payroll tax reforms with linkage (A) L1: Posted earnings (w). (B) L1: labour cost (z). (C) L2: Posted earnings (w) and (D) L2: Labour cost (z)

Notes: The figure shows the evolution of average real posted earnings (A and C) and average real labour cost (B and D) for reforms pertaining to payroll taxes with tax-benefit linkage (L1 and L2), separately for workers in the treatment and control groups. For Reform L1 (A and B), the treatment (control) group consists of workers whose posted earnings in the reference year (1999) were between 1 and 1.4 (0.9 and 1) times the SST that year, using an unbalanced panel of workers who are observed in the reference year and at least another year. For Reform L2 (C and D), the treatment (control) group consists of workers whose posted earnings in the reference year (1989) were between 4 and 5.5 (2.5 and 4) times the SST that year, using a balanced panel of workers. Earnings levels are normalized to 100 for each group in the reference year. In each panel, the shaded area indicates the reform's implementation period, while the vertical dotted line marks the reference year. Sources: DADS Panel; TAXIPP 0.4.

(a) Figure 2: earnings responses, linked reforms.

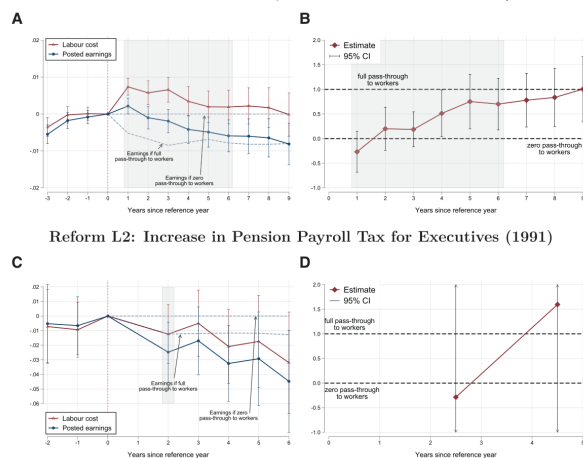


FIGURE 3

Estimated pass-through to workers for payroll tax reforms with linkage (A) L1: DiD estimates. (B) L1: Pass-through to workers (2SLS). (C) L2: DiD estimates and (D) L2: Pass-through to workers (2SLS)

Notes: (A) and (C) show the differential growth in the log of real posted earnings (circle markers) and average real labour costs (triangle markers) between two groups of workers that were affected differently by reforms L1 and L2, while (B) and (D) show the estimated pass-through of the payroll tax increases to workers. For Reform L1 (A and B), the treatment (control) group consists of workers whose posted earnings in the reference year (1999) were between 1 and 1.4 (0.9 and 1) times the SST that year, using an unbalanced panel of workers who are observed in the reference year and at least another year. For Reform L2 (C and D), the treatment (control) group consists of workers whose posted earnings in the reference year (1989) were between 4 and 5.5 (2.5 and 4) times the SST that year, using a balanced panel of workers. In each panel, the shaded area indicates the reform's implementation period, while the vertical dotted line marks the reference year. In (A) and (C), the markers represent the differential growth in posted earnings and labour costs between treated and control workers for both pre-reform and post-reform years, relative to the reference year. The vertical T-bars show the 95% confidence intervals around the point estimates, with standard errors clustered at the individual level. The dashed lines represent the predicted evolution of posted earnings under the counterfactual scenarios of zero pass-through and full pass-through of the payroll tax change to workers. (B) and (D) present estimates of pass-through to workers, obtained from a 2SLS estimation of equation (6) in the main text, where $\log(1 + \tau)$ is instrumented by the interaction between the treatment group and year dummies. Diamond markers show the estimates while the vertical T-bars show the 95% confidence intervals (arrows indicate confidence intervals that extend beyond the y-axis range displayed). In the case of Reform L2, given the limited sample size, we aggregate years 2 and 3, as well as years 4 and 5. Sources: DADS Panel; TAXIPP 0.4.

(b) Figure 3: pass-through estimates, linked reforms.

TABLE 3
Pass-through of payroll tax reforms with tax-benefit linkage

Reform:	Payroll tax reforms			
	L1		L2	
	Increase in pension payroll tax for non-executives		Increase in pension payroll tax for executives	
Reference year (t_0):	1999	1999	1989	1989
Last year ($t_{\max}$):	2008	2008	1995	1995
Side:	Employer	Employer + Employee	Employer	Employer + Employee
	(1)	(2)	(3)	(4)
<i>Pass-through to workers:</i>				
Last year	1.01*** (0.34)	1.00*** (0.19)	1.26 (2.01)	1.20 (1.52)
Final years	0.88*** (0.28)	0.93*** (0.15)	0.74 (1.46)	0.78 (1.17)
N	496,039	496,039	13,240	13,240

Notes: This table reports difference-in-differences estimates of the pass-through to workers of payroll tax reforms with tax-benefit linkage. For Reform L1 (Columns 1 and 2), the treatment (control) group consists of workers whose posted earnings in the reference year (1999) were between 1 and 1.4 (0.9 and 1) times the SST that year, using an unbalanced panel of workers who are observed in the reference year and at least another year. For Reform L2 (Columns 3 and 4), the treatment (control) group consists of workers whose posted earnings in the reference year (1989) were between 4 and 5.5 (2.5 and 4) times the SST that year, using a balanced panel of workers. The pass-through estimates are obtained from a 2SLS regression of log(posted earnings) (Columns 1 and 3) or log(net earnings) (Columns 2 and 4) on $\log(1 + \tau)$ (see equation (6) in the main text), where τ denotes the average employer payroll tax rate (Columns 1 and 3) or the sum of the average employer and employee payroll tax rates (Columns 2 and 4), and $\log(1 + \tau)$ is instrumented by the interaction between the treatment group and year dummies. "Last year" is the pass-through estimate in the last year of the post-reform period, while "Final years" is the average over the last 3 years. Standard errors clustered at the individual level are shown in parentheses. *** $p < 0.01$.
Sources: DADS Panel; TAXIPP 0.4.

Table 3: pass-through estimates for payroll tax reforms with tax-benefit linkage.

3.4 Figures 4–5 and Table 4: unlinked employer-side reforms

Read: Figures 4 and 5:

- Figure 4 shows that unlinked employer payroll tax changes affect labour costs but do not lead to the same earnings adjustment seen for linked reforms.
- Figure 5 shows pass-through estimates for NL1 and NL2.
- NL1 shows little pass-through to workers.
- NL2 is more ambiguous visually because of pre-trends, but it does not support full pass-through.

Read: Table 4, employer-side columns:

- NL1 has last-year pass-through of 0.25 and final-years pass-through of -0.02.
- NL2 has last-year pass-through of 0.44 and final-years pass-through of 0.41, but robustness checks with trends reduce the estimate.
- NL3, the payroll-based tax credit, has estimates close to zero.

Economic message: Without tax-benefit linkage, employer-side taxes remain largely with employers at the directly treated-worker level. This challenges the simple idea that all employer payroll taxes are automatically shifted to workers.

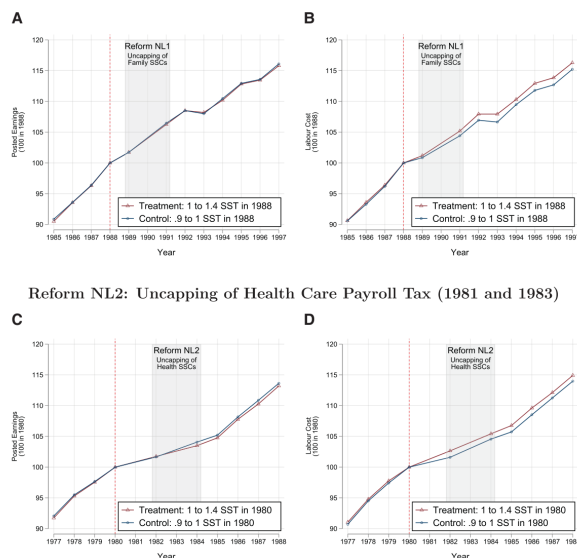


FIGURE 4

Earnings responses to payroll tax reforms without linkage (A) NL1: Posted earnings (w). (B) NL1: Labour cost (z). (C) NL2: Posted earnings (w) and (D) NL2: Labour cost (z)

Notes: The figure shows the evolution of average real posted earnings (A and C) and average real labour cost (B and D) for reforms pertaining to payroll taxes without tax-benefit linkage (NL1 and NL2), separately for workers in the treatment and control groups. The results for Reform NL1 (uncapping of family payroll tax) are shown in (A) and (B), while the results for Reform NL2 (uncapping of health care payroll tax) are shown in (C) and (D). For both reforms, the treatment (control) group consists of workers whose posted earnings in the reference year (1988 for NL1 and 1980 for NL2) were between 1 and 1.4 (0.9 and 1) times the SST that year, using an unbalanced panel of workers who are observed in the reference year and at least another year. Earnings levels are normalized to 100 for each group in the reference year. In each panel, the shaded area indicates the reform's implementation period, while the vertical dotted line marks the reference year. Sources: DADS Panel; TAXIPP 0.4.

(a) Figure 4: earnings responses, unlinked payroll taxes.

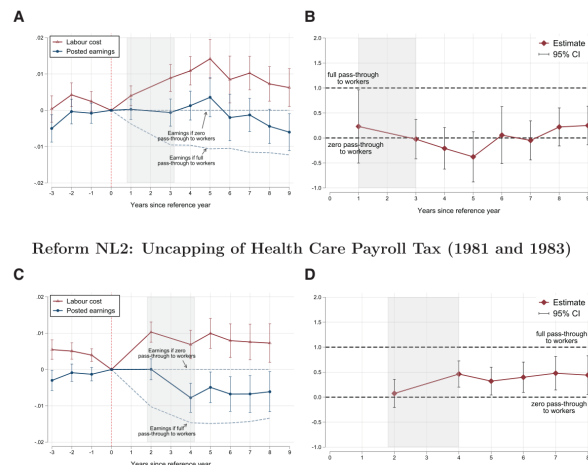


FIGURE 5

Estimated pass-through to workers for payroll tax reforms without linkage (A) NL1: DiD estimates. (B) NL1: Pass-through to workers (2SLS). (C) NL2: DiD estimates and (D) NL2: Pass-through to workers (2SLS)

Notes: (A) and (C) show the estimated log differences in average real posted earnings (circle markers) and average real labour costs (triangle markers) between two groups of workers that were affected differently by reforms NL1 and NL2, while (B) and (D) show the estimated pass-through of the payroll tax increases to workers. The results for Reform NL1 (uncapping of family payroll tax) are displayed in (A) and (B), while the results for Reform NL2 (uncapping of health care payroll tax) are displayed in (C) and (D). For both reforms, the treatment (control) group consists of workers whose posted earnings in the reference year (1988 for NL1 and 1980 for NL2) were between 1 and 1.4 (0.9 and 1) times the SST that year, using an unbalanced panel of workers who are observed in the reference year and at least another year. In each panel, the shaded area indicates the reform's implementation period, while the vertical dotted line marks the reference year. In (A) and (C), the markers represent the differential growth in posted earnings and labour costs between treated and control workers for both pre-reform and post-reform years, relative to the reference year. The vertical T-bars show the 95% confidence intervals around the point estimates, with standard errors clustered at the individual level. The dashed lines represent the predicted evolution of posted earnings under the counterfactual scenarios of zero pass-through and full pass-through of the payroll tax change to workers. (C) and (D) present estimates of pass-through to workers, obtained from a 2SLS estimation of equation (6) in the main text, where $\log(1 + \tau)$ is instrumented by the interaction between the treatment group and year dummies. Diamond markers show the estimates while the vertical T-bars show the 95% confidence intervals. Sources: DADS Panel; TAXIPP 0.4.

(b) Figure 5: pass-through, unlinked payroll taxes.

TABLE 4
Pass-through of payroll and income tax reforms without tax-benefit linkage

Reform:	Payroll tax reforms		Income tax reforms	
	NL1 Uncapping of family payroll tax	NL2 Uncapping of health care payroll tax	NL3 Tax credit on employer payroll tax	NL4 Employee income tax on profit-sharing
Reference year (t_0):	1988	1980	2012	1995
Last year ($t_{\max}$):	1997	1988	2018	1999
Side:	Employer (1)	Employer (2)	Employer (3)	Employee (4)
Pass-through to workers:				
Last year	0.25 (0.20)	0.44** (0.20)	-0.06 (0.26)	0.89*** (0.33)
Final years	-0.02 (0.18)	0.41*** (0.13)	-0.05 (0.14)	0.75*** (0.28)
N	594,632	642,289	275,307	555,033

Notes: See notes of Table 3. For Reforms NL1 and NL2 (Columns 1 and 2), the treatment (control) group consists of workers whose posted earnings in the reference year (1988 for NL1 and 1980 for NL2) were between 1 and 1.4 (0.9 and 1) times the SST that year, using an unbalanced panel of workers who are observed in the reference year and at least another year. For Reform NL3 (Column 3), the treatment (control) group consists of workers whose posted earnings in the reference year (2012) were between 2.3 and 2.5 (2.5 and 2.7) times the minimum wage (MW) that year, using an unbalanced panel of workers who are observed in the reference year and at least another year. For Reform NL4 (Column 4), the sample is restricted to workers employed in firms with non-negative profits during the pre-reform period (1993–5) and having at least 10 employees. The treatment group consists of workers in the firms that distributed profit-sharing bonuses at least once in the pre-reform, while the control group consists of workers in the firms that did not distribute such bonuses. For Reform NL4, “Final years” is the average of the pass-through estimates over the last 2 years post-reform. *** $p < 0.01$, ** $p < 0.05$.

Sources: DADS Panel; TAXIPP 0.4.

Table 4: pass-through estimates for reforms without tax-benefit linkage.

3.5 Figures 6–7: income-tax reforms based on payroll

Read:

- Figure 6 shows earnings and labour-cost responses for NL3 and NL4.
- NL3 is employer-side and payroll-based; it has little effect on posted earnings.
- NL4 is employee-side; net earnings fall for treated workers.

- Figure 7 presents the corresponding pass-through estimates.
- The employee-side income tax is largely borne by workers, while the employer-side payroll-based credit is not passed through to workers.

Economic message: The income-tax reforms reinforce the statutory-incidence result in the absence of tax-benefit linkage. The worker-side tax falls on workers; the employer-side tax credit stays with employers at the individual level.

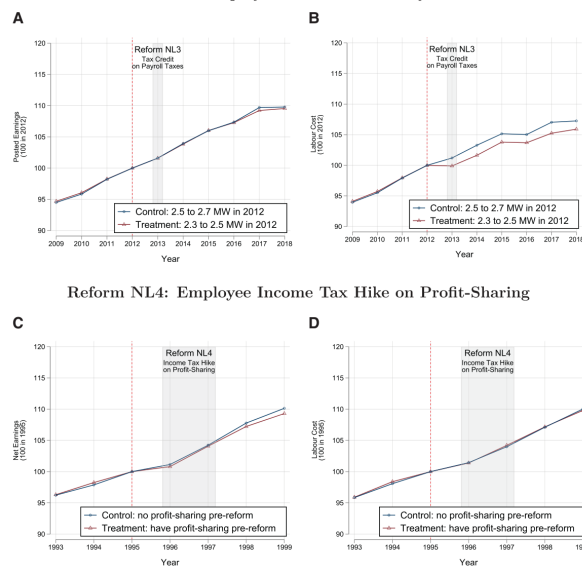


FIGURE 6

Earnings responses to income tax reforms based on payroll (A) NL3: Posted earnings (w). (B) NL3: Labour cost (z). (C) NL4: Net earnings (n) and (D) NL4: Labour cost (z).

Notes: The figure shows the evolution of average real posted earnings (A), real net earnings (C), and average real labour cost (B and D) for income tax reforms without tax-benefit linkage (NL3 and NL4), separately for workers in the treatment and control groups. The results for Reform NL3 (an employer tax credit on payroll taxes for earnings below 2.5 times the minimum wage) are shown in (A) and (B), while the results for Reform NL4 (an increase in employees' flat-rate withholding tax on earnings from profit-sharing) are shown in (C) and (D). For Reform NL3, the treatment (control) group consists of workers whose posted earnings in the reference year (2012) were between 2.3 and 2.5 (2.5 and 2.7) times the minimum wage (MW) that year, using an unbalanced panel of workers who are observed in the reference year and at least another year. For Reform NL4, the sample is restricted to workers employed in firms with non-negative profits during the pre-reform period (1993–5) and having at least 10 employees. The treatment group consists of workers in the firms that distributed profit-sharing bonuses at least once in the pre-reform, while the control group consists of workers in the firms that did not distribute such bonuses. Earnings levels are normalized to 100 for each group in the reference year. In each panel, the shaded area indicates the reform's implementation period, while the vertical dotted line marks the reference year. *Sources:* DADS Panel; TAXIPP 0.4.

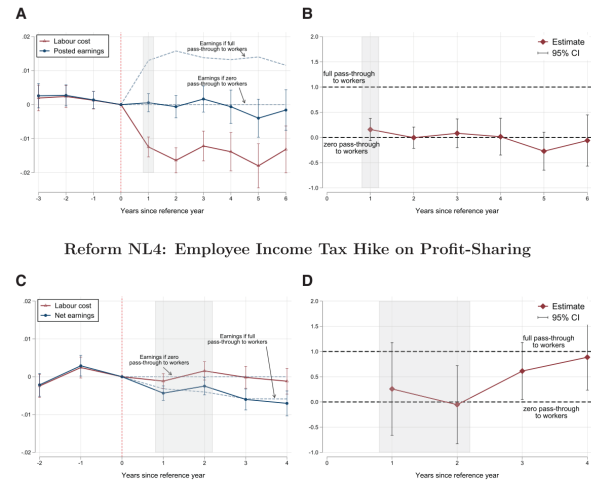


FIGURE 7

Estimated pass-through to workers for income tax reforms based on payroll (A) NL3: DiD estimates. (B) NL3: Pass-through to workers (2SLS). (C) NL4: DiD estimates and (D) NL4: Pass-through to workers (2SLS).

Notes: (A) and (C) show the estimated log differences in the average real posted earnings (A, circle markers), the average real net earnings (C, circle markers), and the average real labour cost (triangle markers) between two groups of workers that were affected differently by reforms NL3 and NL4, while (B) and (D) show the estimated pass-through of the tax increases to workers. The results for Reform NL3 (an employer tax credit on payroll taxes for earnings below 2.5 times the minimum wage) are displayed in (A) and (B), while the results for Reform NL4 (an increase in employees' flat-rate withholding tax on earnings from profit-sharing) are displayed in (C) and (D). For Reform NL3, the treatment (control) group consists of workers whose posted earnings in the reference year (2012) were between 2.3 and 2.5 (2.5 and 2.7) times the minimum wage (MW) that year, using an unbalanced panel of workers who are observed in the reference year and at least another year. For Reform NL4, the sample is restricted to workers employed in firms with non-negative profits during the pre-reform period (1993–5) and having at least 10 employees. The treatment group consists of workers in the firms that distributed profit-sharing bonuses at least once in the pre-reform, while the control group consists of workers in the firms that did not distribute such bonuses. In each panel, the shaded area indicates the reform's implementation period, while the vertical dotted line marks the reference year. In (A) and (C), the markers represent the differential growth in posted earnings (A) or net earnings (C) under the counterfactual scenarios of zero pass-through and full pass-through of the payroll tax change to workers. (B) and (D) present estimates of pass-through to workers, obtained from a 2SLS estimation of equation (6) in the main text, where $\log(1 + \tau)$ is instrumented by the interaction between the treatment group and year dummies. Diamond markers show the estimates while the vertical T-bars show the 95% confidence intervals. *Sources:* DADS Panel; TAXIPP 0.4.

(a) Figure 6: earnings responses, payroll-based income taxes.

(b) Figure 7: pass-through, payroll-based income taxes.

3.6 Table 5 and Figure 8: formal tests and meta-analysis

Read: Table 5:

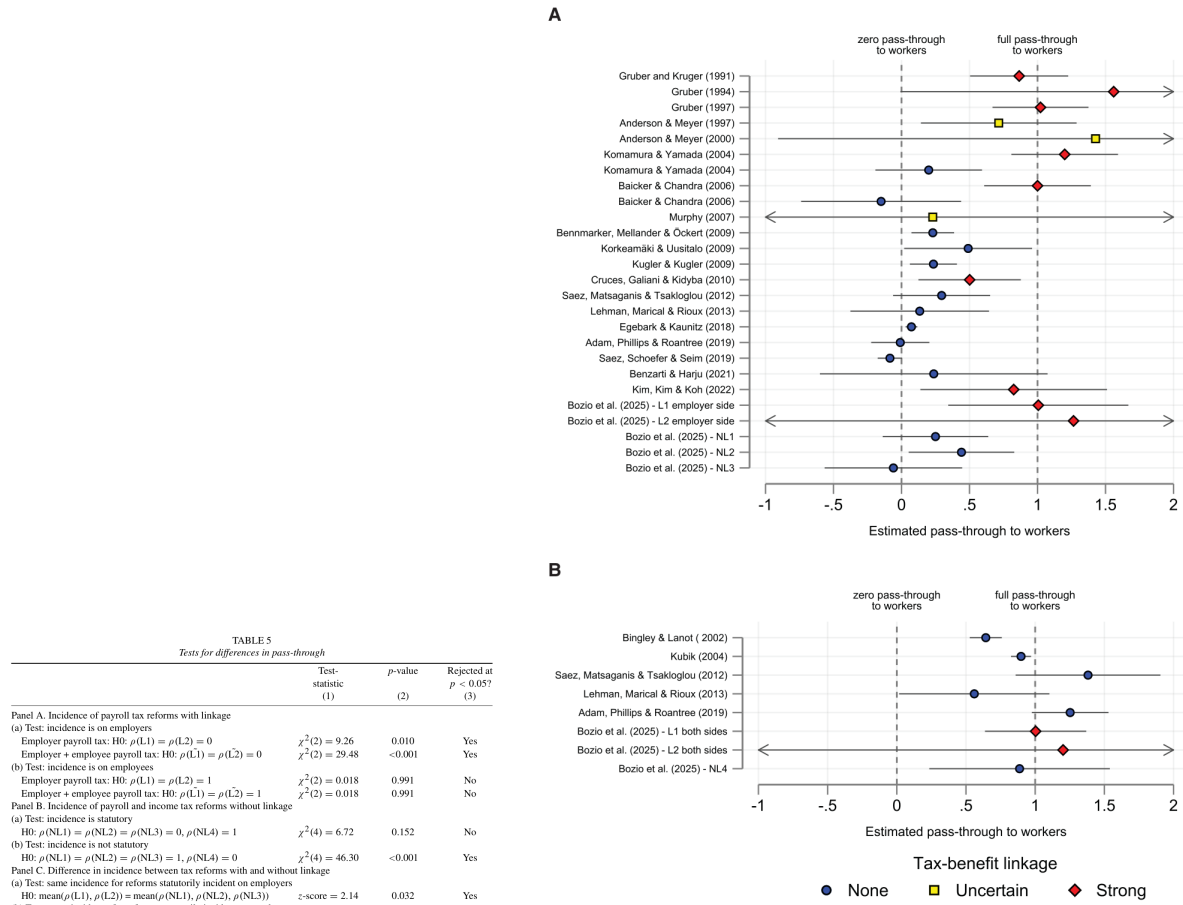
- For linked reforms, the hypothesis of employer incidence is rejected; the hypothesis of worker incidence is not rejected.
- For unlinked reforms, the hypothesis that incidence follows statutory incidence is not rejected.
- The hypothesis of opposite incidence for unlinked reforms is strongly rejected.
- Employer-side linked and unlinked reforms have statistically different average pass-through.

Read: Figure 8:

- Panel A compares employer payroll-tax pass-through estimates across studies.
- Strong-linkage reforms cluster near full pass-through to workers.

- No-linkage reforms cluster near zero or partial pass-through.
- Panel B shows employee-side tax estimates, which are generally near worker incidence.

Economic message: The paper's French estimates fit a broader pattern: tax-benefit linkage explains why some payroll-tax studies find full shifting and others find little shifting.



(a) Table 5: formal tests.

(b) Figure 8: meta-analysis of payroll tax incidence.

3.7 Key equation excerpts from the original paper

Read:

- The first excerpt contains the conceptual pass-through formula linking ρ , σ , η^S , and q .
- The second and third excerpts contain the event-study and 2SLS estimating equations used to recover pass-through.

Economic message: These equations connect the conceptual framework to the empirical design: the theory says what ρ means, and the event-study/2SLS framework estimates it using tax variation around reforms.

exploits a change in the average tax rate for workers above versus below the threshold—except for NL3, where the reform directly altered the average tax rate. The validity of this difference-in-differences (DiD) approach hinges on the assumption that, in the absence of the reform, the average earnings of workers in the treatment and control groups would have followed parallel trends. For the income tax reform on the employee side (NL4), we adopt a different definition of the treatment and control groups, distinguishing workers based on whether their firms distributed profit-sharing bonuses before the reform.

5.1. Baseline specification

To evaluate the incidence of employer payroll taxes on posted earnings (reforms L1, L2, NL1, NL2, and NL3), we rely on a model that is estimated using two-stage least squares (2SLS). For each reform, we estimate the first-stage and reduced-form effects of the payroll tax change using the following panel DiD specification:

$$\log(1 + \tau_{i,t}) = \alpha + \theta_i + \eta_t + \sum_{k=-m}^r \beta_k \cdot (T_i \times \mathbb{1}\{t = t_0 + k\}) + \varepsilon_{i,t}, \quad (4)$$

$$\log w_{i,t} = \mu + \nu_i + \pi_t + \sum_{k=-m}^r \gamma_k \cdot (T_i \times \mathbb{1}\{t = t_0 + k\}) + v_{i,t}. \quad (5)$$

The first-stage equation (4) expresses $\log(1 + \tau_{i,t})$, where $\tau_{i,t} \equiv (z_{i,t}/w_{i,t}) - 1$ is the average employer payroll tax rate that applies to the posted earnings $w_{i,t}$ of worker i in year t , as a function of worker fixed effects (θ_i), year fixed effects (η_t), and the full set of interactions between the treatment group indicator T_i and the year fixed effects, which include m pre-reform years and r post-reform years. The treatment group indicator T_i takes the value 1 if the earnings of worker i imply exposure to the payroll tax reform under consideration (*i.e.* they are above the SST for reforms L1, NL1, and NL2, above four times the SST for reform NL2, and below 2.5 times the minimum wage for NL3), and 0 otherwise. The coefficients β_k on the interaction terms are normalized to zero in the reference year (t_0), which corresponds to the last available pre-reform year. Each coefficient measures the post-reform log difference in average employer payroll tax rates between treated and control workers in year k , relative to the reference year.

The reduced-form equation (5) expresses the log of posted earnings, $\log(w_{i,t})$, as a function of the same variables. Under the assumption that the labour cost trends would have remained parallel for all years $k \geq 1$ in the absence of reform, the coefficients γ_k for $k \geq 1$ capture the reform's impact on posted earnings after k years.¹⁰

Consistent with equation (1), the estimated pass-through of employer payroll taxes to workers k years after the reform, denoted by ρ_k , is given by the opposite of the ratio between the

We further denote by $\eta_k^s \equiv \frac{d \ln \frac{w_t}{w_0}}{d \ln \omega_t}$ the labour supply elasticity of workers of type k with respect to the posted wage.

2.3. Pass-through of payroll taxes to wages

The pass-through rate of payroll taxes to the wage of treated workers relative to control workers, *i.e.* the proportional change in the relative wage of treated workers resulting from a one percent change in the rate of payroll taxes levied on these workers, is denoted by ρ . This measure can be understood as pass-through at the individual level in the sense that treated workers are those directly affected by the change in payroll taxes. Within the above framework, one can derive a general closed-form formula for pass-through (see Appendix A). Assuming that treated and control workers have similar labour supply elasticities (*i.e.* $\eta_T^s \approx \eta_C^s \equiv \eta^s$), the formula simplifies to

$$\rho = -\frac{d \ln \left(\frac{w_T}{w_C} \right)}{d \ln (1 + \tau_T)} \approx \frac{\sigma + \eta^s \cdot q}{\sigma + \eta^s}. \quad (1)$$

(a) Conceptual pass-through formula.

(b) Empirical event-study specifications.

This parameter $\hat{\rho}_k$ can be recovered by estimating the following equation separately for each post-reform year $k \geq 1$ using 2SLS:

$$\log w_{i,t} = \lambda + \phi_i + \omega_t - \rho_k \cdot \log(1 + \tau_{i,t}) + \sum_{\substack{l=-m \\ l \neq k}}^r \delta_l \cdot (T_i \times \mathbb{1}\{t = t_0 + l\}) + \zeta_{i,t}, \quad (6)$$

where ϕ_i and ω_t denote worker and year fixed effects, and the interaction term $T_i \times \mathbb{1}\{t = t_0 + k\}$ is used as an instrument for $\log(1 + \tau_{i,t})$. This procedure ensures that the earnings response in any post-reform year is appropriately scaled by the size of the first-stage effect specific to that year. Indeed, the first-stage and reduced-form equations of this 2SLS model directly correspond

2SLS pass-through specification.

4 Short summary

- The paper studies whether payroll-tax incidence depends on tax-benefit linkage.
- It uses six large French reforms: two linked payroll tax reforms, two unlinked employer payroll tax reforms, and two payroll-like income-tax reforms without linkage.
- The conceptual framework adds a linkage parameter q to the standard labour-market incidence model.
- If $q = 1$, linked payroll taxes are fully shifted to workers regardless of labour supply or substitution elasticities.
- The empirical design compares workers around thresholds before and after reforms using worker and year fixed effects.
- Pass-through is estimated as the ratio of earnings responses to tax changes, implemented through 2SLS.
- Linked pension payroll tax reforms are fully passed through to workers.

- Unlinked employer payroll tax reforms show little pass-through to directly affected workers.
- A payroll-based employer tax credit does not significantly raise worker earnings.
- An employee-side income tax increase is borne by workers.
- Formal tests support full worker incidence for linked reforms and statutory incidence for unlinked reforms.
- Robustness checks address alternative treatment windows, worker-specific trends, doughnut holes, non-parametric evidence, and placebo reforms.
- The meta-analysis shows that linked employer payroll taxes in the literature tend to have pass-through near 100 percent, while unlinked employer payroll taxes tend to have low pass-through.
- The results support the efficiency argument for transparent contribution-benefit linkage in social insurance finance.
- The main caveat is that the paper estimates individual-level incidence; employer-side taxes without linkage might still be shifted at the firm or market level.

5 Conclusion

5.1 Core conclusion

Bozio, Breda, Grenet, and Guillouzouic show that payroll-tax incidence depends critically on tax-benefit linkage. When workers perceive a strong and transparent link between contributions and future benefits, payroll taxes are fully borne by workers through lower earnings. When employer payroll taxes lack such linkage, the directly affected workers do not bear the tax through lower relative earnings; the individual-level incidence is much closer to statutory incidence.

5.2 Economic intuition

Linked payroll taxes operate like deferred wages. A worker who pays more today receives more pension benefits in the future, so the tax is not perceived as a pure wedge. Firms can reduce posted wages without reducing the worker's perceived compensation. By contrast, an unlinked employer payroll tax is perceived as a firm-side cost. Fairness norms, bargaining, or limited individual-level wage adjustment may prevent firms from lowering only the affected workers' wages.

5.3 Takeaway

For policy analysis, the lesson is simple but important: do not treat all payroll taxes as identical. Whether a payroll tax is linked to future benefits can change its incidence, its distributional effects, and its efficiency costs. Clear tax-benefit linkage can make social-insurance contributions more like deferred compensation; absent linkage, statutory incidence appears to matter much more at the worker level.